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**Q1.**

(a) What is the hexadecimal representation of: `0b1111 1110 1110 1101`?

(b) A 32-bit computer has 512 words in its memory. How many individual bytes are there, and what is the required address bus width for this memory?

(c) What is the value of: `(int8_t) ('z' + 42)`?
*Hint: This question is broken down into smaller steps in the 'extra' section*

**Q2.**

Considering number representation:

(a) What is the 4-bit two's complement representation of $-5$?

(b) If you treat them as regular 4-bit binary numbers (i.e. ignore the fact that -5 is encoded in two's complement as per above), what is the outcome if you add $7 + (-5)$?

(c) What about $5 + (-5)$?

(d) And $(-5) + 7$?

**Q3.**

A bar in town is using a new automatic ordering system, where if you display your drinks order as a *sequence* of lights you get 10% off your bill. You have an array of three LEDs, each of which are able to be set to either (R)ed, (G)reen, or (B)lue.

(a) How many possible patterns are there within this sequence?

(b) If the first LED is used to encode between the (R) Drinks, (G) Meals, and (B) Desserts, how many desserts are able to be encoded?

(c) A new head chef has decided that there needs to be 40 different options for meals. Assuming the type-coding in part b) remains, provide two options for implementing this, one where the number of LEDs stays the same, and one where the number of possible colours stays the same.

(d) Remember that information is encoded within specific What beer is encoded within: (R) (R) (G)?

| Code | Meaning |
|---|---|
| RRR | Weiss |
| RRG | Stout |
| RRB | Pilsner |
| RGR | IPA |
| RGG | Porter |
| RGB | Lager |
| RBR | Munchel |
| RBG | Session |
| RBB | APA |

**Q4.**

## Extra

(a) What is the value of: `(int8_t) ('z' + 42)`?
Let's break it down: The computer will:

- Make a sequence of 1s and 0s to represent `'z'`,
- Make a sequence of 1s and 0s to represent 42,
- Add them together, and
- Interpret this resulting sequence of 1s and 0s (i.e. cast) as a **signed** 8-bit integer.

(a) In either binary or hexadecimal notation, how will the computer represent the character `'z'`?

(b) In binary notation, how will the computer represent the number 42?

(c) What is the result of adding (a) and (b) together?
*Hint: This is most easily done in binary column-by-column addition.*

(d) How are all signed (i.e. positive or negative) integers stored in computers?

(e) What is the value therefore stored in the sequence of 1s and 0s from (c), interpreted as an 8-bit signed integer?

(f) Extra for experts: What is the 32-bit floating point representation of -23.25?



**END OF QUESTIONS**